

Exercise 1 Root finding

Suppose f is a continuous function and $f(a) < 0$ and $f(b) > 0$ for some known a and b . For

simplicity, assume $a < b$. Then, there must exist some c such that $f(c) = 0$.

(a) Write a function `root(f, a, b)` that takes a function f and two floats a and b and returns the root c .

Hint: check the sign at the midpoint of the interval.

(b) Remove the assumption that $a < b$, and that $f(a) < 0$ and $f(b) > 0$, if your current code relies on them.

(c) Add a check that prints 'function evals have same sign'

if $f(a) > 0$ and $f(b) > 0$ or if $f(a) < 0$ and $f(b) < 0$.

(Optional) Exercise 2: Recursive root finding

Write a recursive function that finds the root of a function in Exercise 1.